Overcoming Visual Clutter in Vision Language Action Models via Concept-Gated Visual Distillation

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***Abstract*— Vision-Language-Action (VLA) models demonstrate impressive zero-shot generalization but degrade sharply in cluttered scenes, a failure we term the "Precision-Reasoning Gap" and hypothesize to reflect background-induced feature dilution: semantically confusable clutter corrupting the visual grounding required for precise manipulation. We propose Concept-Gated Visual Distillation (CGVD), a training-free, model-agnostic inference-time framework that parses the instruction into a safe set, applies a supplied distractor vocabulary with a two-layer target refinement (cross-validation, spatial disambiguation) that penalizes false positives, and inpaints the remaining distractor regions, preserving scene context outside the edits. Compared against the un-augmented π_0 and GR00T policies in SimplerEnv, CGVD substantially improves robustness to semantic clutter—on *put spoon on towel* with 18 semantic distractors, π_0's success rate rises from 43.0% to 77.5% (Table III)—and, in one exploratory 5-seed condition with compositional prompts, improves attribute adherence. At the same condition, a vocabulary-matched reimplementation of the run-time intervention baseline BYOVLA reaches 32.5% at $\approx 30\times$ CGVD's total per-chunk cost—a dense-clutter regime outside BYOVLA's design envelope (§IV-E). The protection is task- and distractor-type-conditional: on *put carrot on plate*, where contextual clutter aids the baseline policies, CGVD provides no benefit at any tested density and often significantly underperforms them. Runtime compositing adds a measured ≈ 16 ms per frame ($\approx +10\%$ episode wall time) after a one-off per-episode initialization of 0.7–0.9 s. These results support inference-time visual distillation as a practical defense against semantically confusable clutter for frozen VLA policies in static scenes with a known clutter vocabulary.**

I. INTRODUCTION

Vision-Language-Action (VLA) models [1]–[5] ground large language models into robotic control, enabling robots to follow open-vocabulary instructions like "Put spoon on towel" without task-specific training.

However, a significant gap remains between the semantic reasoning capabilities of these models and their geometric precision in deployment conditions. While VLAs excel in curated, clutter-free environments, their performance degrades precipitously in the presence of visual clutter [6], and more broadly under physical variations of the scene [7]. We term this phenomenon the Precision-Reasoning Gap: the model successfully identifies the target object conceptually, yet fails to manipulate it precisely. We hypothesize that attention corruption from surrounding distractors dilutes the latent representation used for spatial planning; under this hypothesis, feature dilution would manifest as high-variance trajectories and manipulation failure. Our experiments (§IV) measure the end-to-end effect of removing distractors; they do not directly instrument the attention mechanism, so we present this account as a working hypothesis rather than an established mechanism.

This degradation is not uniform across distractor types: failure concentrates around distractors sharing visual or semantic properties with the target, consistent with prior reports [6]. While VLAs may exhibit resilience to arbitrary clutter via large-scale pre-training, they remain brittle to semantically confusable objects; for example, a fork scattered near a target spoon triggers conflicting visual tokens within the same affordance category, causing the policy to attend to or even grasp the wrong object.

Existing mitigations fall into three categories (§II): observation-grounding methods such as OBEYED-VLA [8] augment the VLA with a trained perception module that produces task-conditioned, object-centric, geometry-grounded observations, fine-tuning the policy on clean demonstrations; inference-time interventions such as BYOVLA [9] identify distractors with a VLM (an external GPT-4o dependency) and probe per-region sensitivity with multiple VLA forward passes, with threshold-based target protection; and training-time augmentation [10]–[12] generates cluttered training data at the cost of retraining, without deployment-time guarantees.

To bridge this gap without retraining, we propose Concept-Gated Visual Distillation (CGVD): a model-agnostic inference framework leveraging vision foundation models [13] to restructure observations before they reach the policy. CGVD parses the instruction into target and anchor, segments distractors and task-relevant entities independently, and uses set-theoretic subtraction to exclude the target from the distractor mask *conditional on correct safe-set segmentation*: no pixel segmented as safe is inpainted, but an object the segmenter misses is unprotected (§IV-D). Distractors are then replaced via inpainting [14], preserving the surrounding scene context.

Our contributions are as follows:

- **Concept-Gated Visual Distillation (CGVD):** a training-free, model-agnostic inference framework removing distractors from VLA observations via language-grounded segmentation and inpainting while preserving scene context.
- **Interaction-Aware Masking Logic:** a set-theoretic cross-validation pipeline that overcomes the semantic confusion of open-set segmenters (which score prompts independently), penalizing false positives and spatially disambiguating true targets from confusable distractors.

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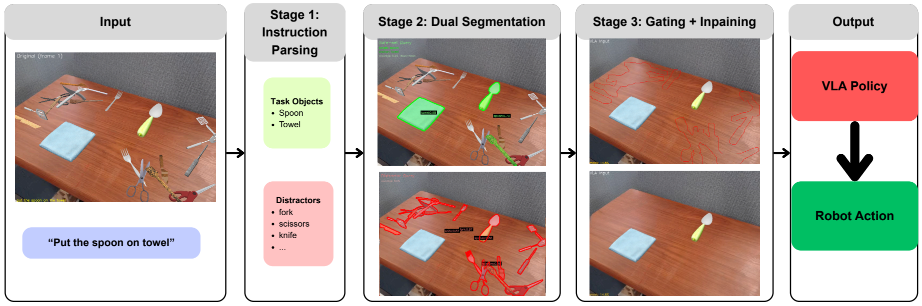

Fig. 1. The CGVD pipeline. **Stage 1:** the instruction is parsed into a safe set (target, anchor); the distractor vocabulary is supplied. **Stage 2:** SAM 3 segments both sets independently. **Stage 3:** set-theoretic gating subtracts the safe-set mask from the distractor mask; LaMa inpaints the result into the distilled observation passed to the VLA.

- **Systematic clutter-density evaluation:** We evaluate CGVD on two VLAs (π_0, GR00T) within the SimplerEnv benchmark over more than 20,000 episodes, comparing each policy with and without CGVD under matched seeds. CGVD prevents policy collapse under dense *semantic* clutter—reaching 77.5% success against the un-augmented policy's 43.0% (π_0, *put spoon on towel*, 18 semantic distractors; Table III)—and improves attribute adherence in an exploratory compositional-prompt study, while *underperforming* the baseline where contextual clutter aids the policy (§IV-B). Comparisons are against the un-augmented policies, plus a measured head-to-head against a BYOVLA [9] reimplementation (§IV-E).

II. Related Work

A. Vision-Language-Action Models

VLA models such as RT-2 [2], OpenVLA [1], and Octo [3] leverage internet-scale pre-training for zero-shot generalization, yet struggle with high-precision geometric grounding in clutter [6]. We hypothesize that the static patch resolution of ViT backbones [15] and the absence of coarse-to-fine zooming [16] make such architectures susceptible to feature dilution [6]: background clutter competing with task-relevant signals for representational capacity.

B. Vision Foundation Models for Robotic Perception

Vision Foundation Models such as SAM 3 [13] and GroundingDINO [17] localize objects from free-form text; robotics has mostly used them to *add* information—value maps [18], semantic maps [19], labeling [20], visual prompting [21]. We invert the paradigm, suppressing task-irrelevant regions: an information bottleneck only in an informal, analogical sense [22] (a hard pixel-space mask, no learned rate–distortion trade-off).

C. Robustness via Training-Time Augmentation

Domain randomization [23] and large-scale pre-training [24], [25] have not made VLAs robust to cluttered-scene shifts [26], [27]. Training-time augmentation generates cluttered or counterfactual data: ROSIE [10], GenAug [11], and NICE [12], whose pixel-space *scene surgery* is the closest prior to our core operation. CGVD performs a related pixel-space edit at *inference* time on a frozen policy: no retraining, instruction-adaptive, but inheriting run-time perception risk.

D. Inference-Time Attention Intervention

Distracting Token Pruning (DTP) [28] suppresses text-misaligned visual tokens in feature space; we conjecture such pruning struggles once distractor and target tokens are entangled by early self-attention, and instead intervene in pixel space, before tokenization. Unlike Visual Prompting [21], which adds information, we remove it. Two prior methods also operate on the observation: OBEYED-VLA [8] pairs the VLA with a *trained* perception module (task-conditioned, object-centric, geometry-grounded views) and fine-tunes the policy on clean single-object demonstrations—CGVD trains and fine-tunes nothing; BYOVLA [9] intervenes at run time, and §IV-E reports a measured comparison.

III. Methodology

CGVD is a training-free, inference-time perception wrapper around any VLA policy (Fig. 1): given instruction l and observation $o_t \in \mathbb{R}^{H\times W\times 3}$, a distillation function ϕ produces $\hat{o}_t = \phi(o_t, l)$ with distractors replaced by background, and the policy acts on $a_t = \pi(\hat{o}_t, l)$. Since ϕ touches only the observation interface, it is architecture-agnostic. The key insight: the instruction already specifies which objects must remain visible, and CGVD uses this signal to gate visual content; the clutter vocabulary is supplied separately (§III-A).

A. Concept-Gated Decomposition

CGVD begins by parsing the language instruction l to extract a target concept c_{tgt} and an anchor concept c_{anc}. For instance, "put spoon on towel" yields $c_{\text{tgt}} = \texttt{spoon}$ and $c_{\text{anc}} = \texttt{towel}$. These concepts define two sets:

- The **safe set** $\mathscr{S} = \{c_{\text{tgt}}, c_{\text{anc}}\}$: entities that must remain visible. The robot arm is also always preserved, but it is handled through a separate proprioceptive channel (§III-E–III-F) rather than through the segmentation-based safe-set mask.
- The **distractor set** $\mathscr{D} = \{d_1, \ldots, d_K\}$: semantic categories to be treated as clutter (e.g., spatula, fork, knife).

The safe set is derived deterministically from the instruction. The distractor set is *not*: it is an externally supplied vocabulary. In our experiments, $\mathscr{D}$ is re-derived at every episode reset from the category labels of the objects the evaluation protocol actually spawned in that episode (§IV-A)—i.e., CGVD receives the exact ground-truth clutter inventory, per episode. At deployment, $\mathscr{D}$ must come from the integrator (a site vocabulary) or an external model. Where prior work uses an LLM to determine relevance [9], our parsing is deterministic and needs no API *in this closed-vocabulary setting*—a different point on a coverage-versus-cost trade-off, not a strict improvement. §IV-D measures this dependence directly at the tested operating point.

B. Text-Prompted Instance Segmentation

Both the safe set and distractor set are segmented from the observation using SAM 3 [13]. The two sets produce independent mask channels:

$$M_{\text{dist}} = \textstyle\bigcup_{d_k \in \mathscr{D}} \textsc{Seg}(o_t, d_k), \tag{1}$$

$$M_{\text{safe}} = \textsc{Seg}(o_t, c_{\text{tgt}}) \cup \textsc{Seg}(o_t, c_{\text{anc}}), \tag{2}$$

where $\textsc{Seg}(o_t, c)$ denotes the union of all instance masks returned by SAM 3 for concept c in observation o_t. The vision encoder runs once on the initialization frame (t=0); its masks are reused across the episode.

C. Two-Layer Target Refinement

Open-set segmenters evaluate text prompts independently, so single-pass detection suffers semantic confusion: a spatula can yield high confidence for "spoon." To ensure the true target survives distillation, we apply a two-layer refinement on the initial frame (t=0).

Layer 1: Cross-Validation. Each target instance s_i receives a genuineness score—the confidence differential between its safe-set and distractor identities:

$$g(s_i) = \sigma_{\text{safe}}(s_i) - \max_{\substack{d_j \in \mathscr{D} \\ \text{IoU}(s_i, d_j) > \eta}} \sigma_{\text{dist}}(d_j), \tag{3}$$

where σ_{safe} and σ_{dist} are object confidences, and η is an IoU threshold ($\eta = 0.30$). When no distractor satisfies $\text{IoU}(s_i, d_j) > \eta$—the common case for an isolated target—the maximum is defined as 0, so $g(s_i)$ reduces to $\sigma_{\text{safe}}(s_i)$. Genuine targets yield $g > 0$, while imposters yield $g < 0$. Negative values are explicitly preserved to actively penalize false positives in the next layer.

Layer 2: Spatial Disambiguation. Even after cross-validation, the target mask may contain fragmented artifacts or multiple disjoint physical objects. To isolate the correct entity, we evaluate each connected component C_k using a composite score:

$$\text{score}(C_k) = \big(1 + g^*(C_k)\big) \cdot \sigma^*(C_k). \tag{4}$$

Here, $g^*(C_k)$ is maximum genuineness, and $\sigma^*(C_k)$ is peak safe-set confidence. These factors jointly favor genuine and high-confidence components. Only the top-scoring component is retained; the pipeline therefore assumes a *single* target instance per scene (§V).

D. Concept-Gated Mask Composition

The refined masks are combined into a final inpainting mask via set-theoretic gating:

$$M_{\text{inp}} = \text{dilate}(M_{\text{dist}}, r_d) \setminus \text{dilate}(M_{\text{safe}}, r_s), \tag{5}$$

where r_d is a distractor dilation radius, and $r_s \geq r_d$ is a safe-set dilation radius that creates a protective buffer; the implementation enforces $r_s \geq r_d$ by construction, and both are 11 px in our configuration. All masks are binarized with a threshold of 0.5 before dilation to eliminate soft-value artifacts.

E. Clean Scene Generation via Inpainting

We generate a single clean scene—an image of the workspace with distractors removed—by applying LaMa [14], a Fourier convolution-based inpainting model, to the initial frame. The inpainting mask is constructed as:

$$M_{\text{lama}} = M_{\text{inp}} \cup \text{dilate}(M_{\text{robot}}, r_e), \tag{6}$$

where M_{robot} is the robot mask (obtained as described in §III-F) and r_e is a robot dilation radius (20 px). Inpainting the robot region synthesizes the background *behind* the arm, leaving no stale robot pixels once it moves. A consequence is that scene content occluded by the arm at t=0 is synthesized rather than observed for the remainder of the episode (§V). LaMa fills the masked regions with photorealistic background texture. The clean scene is computed once per episode and cached for all subsequent frames.

F. Temporally Consistent Compositing

At each timestep $t > 0$, the distilled observation is produced by alpha-blending the live camera frame with the cached clean scene:

$$\hat{o}_t = \alpha \odot \hat{o}_{\text{clean}} + (1 - \alpha) \odot o_t, \tag{7}$$

where the compositing mask $\alpha \in [0,1]^{H \times W}$ is computed *once* at t=0 and held fixed: the binarized, safe-set-gated distractor mask is feathered with a Gaussian blur ($\sigma = 3.0$), clamped to 1 in core distractor regions and 0 in the (dilated) target region. So that inpainting artifacts never obscure the arm, robot pixels are overwritten onto the composite every step—the only per-frame operation. In simulation this robot mask comes from SimplerEnv's renderer segmentation buffer (per-link actor IDs): pixel-perfect, jitter-free, and *simulation-privileged*; §IV-D measures the cost of substituting a per-frame SAM 3 prediction; on hardware, forward-kinematics projection provides a deterministic, perception-free replacement.

Two properties are explicit. First, $\hat{o}_t$ is the *only* visual input the policy receives. Second, the characteristic failure mode of observation editing is *false-positive erasure*: an object misclassified as a distractor—or entering the workspace after t=0, where the fixed α can blend it toward the cached background—is removed from the policy's view while remaining in the world, and the arm can collide with what it cannot see. Our evaluation is therefore scoped to static workspaces with no human entry, a fixed camera, and a known clutter vocabulary.

IV. Experiments

We evaluate CGVD across two VLA architectures on tabletop manipulation tasks with controlled distractor injection. Our experiments address three questions: (1) Does CGVD improve clutter robustness across different VLA architectures and tasks? (2) How does performance scale with distractor density? (3) How do individual CGVD components contribute at the operating point tested?

A. Experimental Setup

Environment. We evaluate in SimplerEnv [29], whose real-to-sim correlation supports *policy-level* transfer. It does not bound CGVD's perception-level risk: SAM 3 [13] and LaMa [14] were trained on real imagery and are applied here to rendered frames—itself a domain shift—and two pipeline inputs are simulation-privileged (the renderer robot mask, §III-F; the distractor vocabulary, §III-A). All conclusions are scoped to simulation (§V).

All experiments use the WidowX robotic arm with a single fixed third-person camera, matching the Bridge dataset training setup. We test two Bridge dataset tasks: (1) *put spoon on towel* and (2) *put carrot on plate*.

Policies. π_0 refers to the open-source `open-pi-zero` reproduction of π_0 [5] (`github.com/allenzren/open-pi-zero`), Bridge checkpoint `bridge_beta`; GR00T refers to `nvidia/GR00T-N1.6-bridge` [4].[1] Both are frozen; CGVD touches only their image input.

Distractors. We introduce a three-way distractor taxonomy: *Semantic* (high semantic proximity and shared visual characteristics with the target), *Random* (no similarity), and *Attribute* (same category and form, different physical properties); [6] documents the magnitude of clutter-induced degradation but uses a continuous clutter measure rather than these types. Assets come from RoboCasa [27] and YCB [30], placed collision-aware on a fixed 6 cm grid (one object per cell); an unplaceable object falls back to a collision-free random position or is hidden off-scene—episodes are never dropped. The fallback engages at the highest density: across 1,400 audited re-executions of the headline condition (identical placement generator and seeds), 18.7% of episodes realized fewer than the nominal 18 objects (mean 17.6; ≥ 16 in 96%), making realized-density trends, if anything, conservative. The vocabulary $\mathscr{D}$ is re-derived per episode from the actually-spawned assets' category labels (`rc_fork_0` → "fork"): it coincides exactly with each episode's injected inventory (§III-A).

Protocol. We report success rate (SR) via SimplerEnv's built-in per-task predicate over 60-step episodes: 20 episodes $\times$ 10 seeds per condition (200 episodes; the attribute study of Table II uses 5 seeds), with matched seeds between arms. Seed $s \in \{0,\ldots,9\}$ seeds the RNGs; episode e uses canonical initial layout $(s+e)$ mod 24 and distractor seed $10^4 s + e$, so both arms see identical states and clutter. Counts are sampled at $\{0,4,8,10,14,18\}$; no episodes were excluded. We compare CGVD against the un-augmented policies throughout, and against a reimplementation of BYOVLA [9] at the headline condition (§IV-E). OBEYED-VLA's released code covers its perception module but not the action policy or robot interface, so an end-to-end comparison was infeasible; DTP's [28] only advertised release is an anonymized review-time repository that has since expired.

Analysis. We report episode-level mean success rates with, as dispersion, the SD of the 10 per-seed rates; differences carry 95% CIs from the per-seed paired deltas (t_9). Episode-level clustering by seed is negligible (one-way ICC across all arms: median -0.02, max 0.06). The headline contrast—π_0, *put spoon on towel*, 18 semantic distractors—is the single confirmatory comparison ($\Delta = +34.5$ pp, 95% CI $[+20.8, +48.2]$, paired t $p = 3\times10^{-4}$; Wilcoxon $p = 0.002$); all other contrasts are exploratory and unadjusted, and deltas whose CI contains zero are within run-to-run noise. The 0-distractor cells are shared between the semantic and random conditions (physically identical). The one condition executed twice end-to-end (π_0, *carrot*, random, 18) differed between batches by 3.0 pp (baseline) and 0.5 pp (CGVD), a direct run-to-run replication estimate.

Implementation. All CGVD parameters, held fixed across every reported experiment: SAM 3 checkpoint `facebook/sam3`; LaMa big-lama (`simple-lama-inpainting`); confidence thresholds 0.30 (safe set) and 0.20 (distractors); $\eta = 0.30$; $r_d = r_s = 11$ px; $r_e = 20$ px; compositing $\sigma = 3.0$; mask binarization 0.5; minimum connected component 50 px; one warm-up frame; clean scene cached once per episode, never refreshed; SAM 3 prompts are the instruction's noun phrases and the category labels of $\mathscr{D}$. Values were selected on small pilot rollouts (primarily spoon-task) and frozen before all reported runs (verified from version-control history). Code, configuration, and evaluation scripts are available at `github.com/awd1779/pi0`.

B. Main Results

Figure 2 plots success rate against distractor count (0–18); Table I reports the numbers with per-seed dispersion and paired CIs.

On *put spoon on towel*, baseline performance collapses under semantic clutter and CGVD prevents it: π_0 gains $+14.0$ to $+34.5$ pp for $n \geq 4$, GR00T $+11.0$ to $+13.5$ pp for $n \geq 8$, every paired CI excluding zero. The gap widens with density: the per-seed slope of the CGVD$-$baseline difference is $+1.54$ pp per distractor for π_0 ($t(9)$=5.54, p<0.001) and $+0.74$ for GR00T ($t(9)$=2.77, p=0.02). Random clutter shows the same direction at high density (π_0: $+13.0$ $[+2.0, +24.0]$; GR00T: $+17.0$ $[+7.6, +26.4]$ at 18 distractors).

Conversely, on *put carrot on plate* the baseline policies *improve* at moderate densities before degrading—plausibly because moderate contextual clutter better matches the object-

[1] Ref. [4] documents GR00T N1; the evaluated checkpoint is the N1.6 revision (different backbone and action head) — see the NVIDIA model card.

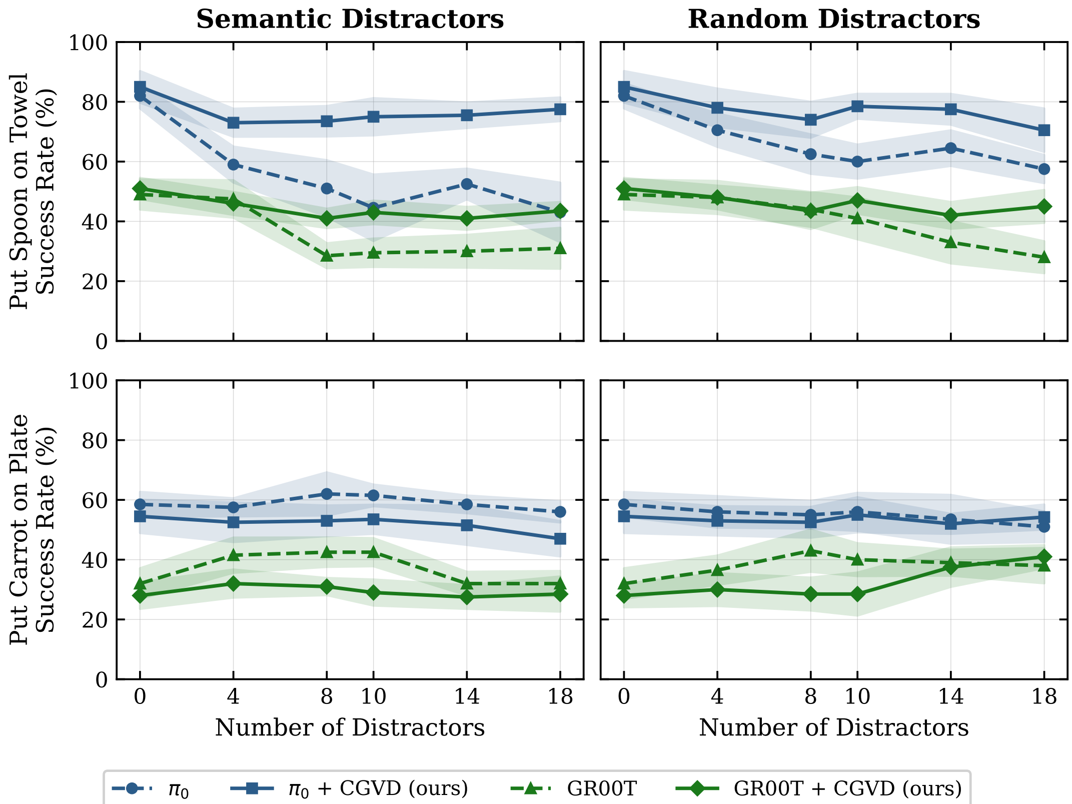

Fig. 2. Success rate vs. distractor count. **Left:** semantic; **right:** random distractors. **Top:** *put spoon on towel*; **bottom:** *put carrot on plate*. Dashed: baseline VLA; solid: +CGVD; colors denote architectures. Each point is the mean over 200 rollouts (20 episodes $\times$ 10 matched seeds) at counts $\{0,4,8,10,14,18\}$ (19,200 episodes total); 0-distractor points are shared between panels. Numeric values with dispersion and CIs: Table I.

TABLE I

MAIN RESULTS (NUMERIC). SUCCESS RATE (%), MEAN $\pm$ SD OVER 10 MATCHED SEEDS (200 EPISODES/CELL); Δ WITH 95% CI FROM PER-SEED PAIRED DELTAS (t_9), BOLD WHEN THE CI EXCLUDES ZERO. DATA OF FIG. 2.

		Semantic distractors						**Random distractors**					
		π_0			GR00T			π_0			GR00T		
Task	**#D**	base	+CGVD	Δ [95% CI]	base	+CGVD	Δ [95% CI]	base	+CGVD	Δ [95% CI]	base	+CGVD	Δ [95% CI]
spoon on towel	0	82.0	85.0	+3.0 [-5.6,+11.6]	49.0	51.0	+2.0 [-6.1,+10.1]	82.0	85.0	+3.0 [-5.6,+11.6]	49.0	51.0	+2.0 [-6.1,+10.1]
	4	59.0	73.0	**+14.0 [+5.4,+22.6]**	47.5	46.0	-1.5 [-8.7,+5.7]	70.5	78.0	+7.5 [-0.8,+15.8]	48.0	48.0	+0.0 [-7.0,+7.0]
	8	51.0	73.5	**+22.5 [+10.7,+34.3]**	28.5	41.0	**+12.5 [+6.4,+18.6]**	62.5	74.0	**+11.5 [+1.7,+21.3]**	44.0	43.5	-0.5 [-12.5,+11.5]
	10	49.0	72.5	**+23.5 [+12.8,+34.2]**	29.5	43.0	**+13.5 [+6.1,+20.9]**	60.0	78.5	**+18.5 [+10.2,+26.8]**	41.0	47.0	+6.0 [-3.0,+15.0]
	14	52.5	75.5	**+23.0 [+14.7,+31.3]**	30.0	41.0	**+11.0 [+3.3,+18.7]**	64.5	77.5	**+13.0 [+1.7,+24.3]**	33.0	42.0	+9.0 [-0.9,+18.9]
	18	43.0	77.5	**+34.5 [+20.8,+48.2]**	31.0	43.5	**+12.5 [+4.0,+21.0]**	57.5	70.5	**+13.0 [+2.0,+24.0]**	28.0	45.0	**+17.0 [+7.6,+26.4]**
carrot on plate	0	58.5	54.5	-4.0 [-11.7,+3.7]	32.0	28.0	-4.0 [-12.2,+4.2]	58.5	54.5	-4.0 [-11.7,+3.7]	32.0	28.0	-4.0 [-12.1,+4.1]
	4	57.5	52.5	-5.0 [-11.5,+1.5]	41.5	32.0	**-9.5 [-12.1,-6.9]**	56.0	53.0	-3.0 [-10.8,+4.8]	36.5	30.0	-6.5 [-16.5,+3.5]
	8	62.0	53.0	-9.0 [-21.1,+3.1]	42.5	31.0	**-11.5 [-17.1,-5.9]**	55.0	52.5	-2.5 [-10.8,+5.8]	43.0	28.5	**-14.5 [-21.1,-7.9]**
	10	61.5	53.5	**-8.0 [-15.0,-1.0]**	42.5	29.0	**-13.5 [-20.7,-6.3]**	56.0	55.0	-1.0 [-10.7,+8.7]	40.0	28.5	**-11.5 [-18.7,-4.3]**
	14	58.5	51.5	-7.0 [-17.4,+3.4]	32.0	27.5	-4.5 [-9.7,+0.7]	53.5	52.0	-1.5 [-13.8,+10.8]	39.0	37.5	-1.5 [-9.2,+6.2]
	18	56.0	47.0	**-9.0 [-17.7,-0.3]**	32.0	28.5	-3.5 [-10.0,+3.0]	49.5	54.0	+4.5 [-6.9,+15.9]	38.0	41.0	+3.0 [-3.6,+9.6]

rich scenes of their pre-training distributions, providing visual anchors.

Here CGVD never significantly outperforms the baseline and significantly underperforms it at several densities: π_0's semantic deltas span -4.0 to -9.0 pp (CIs exclude zero at n=10, 18), and GR00T loses -9.5 to -14.5 pp at moderate densities under both clutter types. When a task benefits from contextual clutter, masking deprives the VLA of useful anchors, and inpainting artifacts can disrupt spatial geometry: the intervention is *task- and distractor-type-conditional*, not universally beneficial.

Qualitative attention visualizations consistent with the working hypothesis (final-query-token attention over the 256 SigLIP tokens, averaged across the PaliGemma backbone's layers and heads) are provided in the code repository. We emphasize that such visualizations are entailed by the intervention itself (removing a distractor's pixels removes attention on them) and that no attention statistic is measured in this paper; alternative explanations—such as CGVD restoring the observation toward the clutter-sparse Bridge training distribution, or simply deleting the wrong-grasp option—are consistent with our data and are not ruled out. Indeed, the mean-color-fill ablation (§IV-D), in which removing the *same* semantic content with a cruder fill forfeits most of the gain,

TABLE II

ATTRIBUTE DISTRACTOR SENSITIVITY ON *put spoon on towel* (π_0, 100 EPISODES/CELL, 5 MATCHED SEEDS). Δ: 95% CI FROM PER-SEED PAIRED DELTAS; BOLD = CI EXCLUDES ZERO; OTHERWISE WITHIN RUN-TO-RUN NOISE.

# Distractors	**Simple Prompt**			**Complex Prompt**		
	π_0	**+CGVD**	**Δ [95% CI]**	π_0	**+CGVD**	**Δ [95% CI]**
0	87.0	85.0	-2.0 [-13.3,+9.3]	86.0	87.0	+1.0 [-5.8,+7.8]
1	79.0	80.0	+1.0 [-7.1,+9.1]	74.0	69.0	-5.0 [-26.9,+16.9]
2	76.0	79.0	+3.0 [-14.9,+20.9]	69.0	77.0	+8.0 [-9.9,+25.9]
3	77.0	75.0	-2.0 [-20.4,+16.4]	64.0	74.0	+10.0 [-7.6,+27.6]
4	75.0	70.0	-5.0 [-24.6,+14.6]	57.0	73.0	**+16.0 [+1.2,+30.8]**

indicates that fill appearance, not distractor semantics alone, carries a large share of the effect.

C. Fine-Grained Semantic Grounding

The precision-reasoning gap is most acute when distractors share the target's semantic class but differ in attributes: complex queries like "Put spoon with green handle on towel" are often reduced to a bag-of-words, entangling the target with fully green objects or dropping modifiers. We use the *Attribute* type (§IV-A): a green-handled spoon target among same-category spoons differing in color or material. We evaluate the baseline and CGVD across 0–4 attribute distractors using two prompt structures: Simple ("Put green spoon on towel") and Complex ("Put spoon with green handle on towel"). CGVD receives the same instruction as the policy; its SAM 3 target prompt is the instruction's attribute phrase ("green spoon" / "spoon with green handle"). Coverage note: the attribute type is evaluated only with π_0, one task, 0–4 distractors (vs. Fig. 2's full grid); attribute claims are scoped accordingly.

Table II reveals a two-sided dependence on descriptive density. The baseline degrades mildly with clutter under simple adjective-noun prompts (87.0% $\rightarrow$ 75.0%) but collapses on compositional queries (86.0% $\rightarrow$ 57.0%)—the bag-of-words brittleness. For CGVD, the complex prompt is what unlocks the benefit: it holds a stable floor under the compositional query (73.0% at 4 distractors, $\Delta = +16.0$, the one delta whose CI excludes zero) because SAM 3 exploits the rich attribute phrase to isolate the correct instance, whereas under the *simple* prompt CGVD provides no reliable benefit (deltas -5.0 to $+3.0$, all within noise): "green spoon" gives the segmenter too little signal to separate the target from attribute-conflicting spoons. Both arms of this study were re-executed for this version under the instruction-consistent protocol above: in the earlier version's runs the segmenter received the scene-informed phrase "spoon with green handle" under both prompt structures—knowledge not derivable from the simple instruction—which restores large gains (87.0/82.0/83.0 at 2–4 distractors vs. 79.0–70.0); removing this asymmetry lowered CGVD's simple-prompt numbers, a correction against our own interest. The 5-seed sample matches the study's original protocol, and per-cell values here are descriptive. CGVD converts descriptive density into robustness; it does not rescue a prompt that lacks it.

TABLE III

ABLATION STUDY (π_0, *put spoon on towel*, 18 SEMANTIC DISTRACTORS; 200 EPISODES/ROW). ROWS REMOVE OR SUBSTITUTE ONE COMPONENT. SR: MEAN $\pm$ SD OVER PER-SEED RATES; Δ VS. FULL PIPELINE WITH 95% CI FROM PAIRED DELTAS. BASELINE/FULL ROWS REUSE THE FIG. 2 RUNS; THE ROBOT-MASK EFFECT IS NOT DISTINGUISHABLE FROM ZERO.

Configuration	**SR (%)**	**Δ [95% CI]**
Baseline (no CGVD)	43.0 ± 16.0	-34.5 $[-48.2, -20.8]$
CGVD (full pipeline)	$\mathbf{77.5} \pm 6.3$	—
LaMa $\rightarrow$ mean-color fill	56.5 ± 16.0	-21.0 $[-35.3, -6.7]$
– Two-layer target refinement	65.0 ± 8.8	-12.5 $[-20.3, -4.7]$
– Robot mask protection	73.0 ± 7.1	-4.5 $[-10.4, +1.4]$

D. Ablation Studies

We ablate the pipeline at a single operating point—*put spoon on towel*, π_0, 18 semantic distractors (Table III); π_0's higher success rates provide headroom to resolve component contributions. The attributions are claims about this operating point only: §IV-B shows the pipeline's net effect reverses in other regimes.

Removing the Two-Layer Target Refinement reduces SR from 77.5% to 65.0%: without cross-validation, genuine targets are erroneously inpainted out of the scene—CGVD's own protection fails, which is why we call it conditional (§III-A). To quantify the *full* pipeline's residual failure rate, we audited a fresh 200-episode re-execution of the headline condition against the renderer's ground-truth target mask: the inpainting mask overlapped the target at $t{=}0$ in 4/200 episodes (2.0%), all four essentially complete erasures. The protection is therefore strong but probabilistic in practice, differing from sensitivity-probe approaches in failure *rate*, not failure *mode*. (This re-execution also provides a replication of the headline cell: 76.0% vs. the original 77.5%.) Substituting a mean-color fill for LaMa incurs the largest single drop (56.5%) despite identical masks and removed content; we hypothesize (unmeasured) that the stark boundaries act like out-of-distribution patches to the ViT backbone. Notably, a majority of the gain at this operating point thus depends on *how* regions are filled, not on removal alone. Finally, removing Robot Mask Protection reduces SR to 73.0%; this -4.5 pp effect is not statistically distinguishable from zero at our sample size (95% CI $[-10.4, +1.4]$), so we treat robot-mask protection as qualitatively motivated—without it, the compositing mask occasionally and visibly occludes the robot arm—rather than as a measured gain.

Deployable robot mask. To measure the cost of the renderer mask's simulation privilege, we additionally ran the full pipeline with a *per-frame SAM 3 prediction* (prompt "robot arm") replacing the renderer mask, on the same seeds. Success is statistically unchanged—74.0% vs. 76.0% for a ground-truth-mask re-execution (paired $\Delta = -2.0$ pp, 95% CI $[-11.3, +7.3]$)—but the deployable path costs 240.7 ± 1.9 ms per frame ($\approx$0.96 s per 4-step chunk) and SAM 3 returns *no* robot detection in 20.1% of the 12,400 frames, falling back to the previous mask—leaving the arm's leading edge unprotected over formerly-inpainted regions during such frames, with no bound on staleness. No accuracy difference was detected ($\Delta = -2.0$ pp), though the experiment only excludes accuracy costs larger than $\approx$11 pp; the measured privilege is latency and reliability, and forward-kinematics projection (§III-F)—which avoids this failure class entirely—remains the sensible hardware path.

Vocabulary sensitivity. Since $\mathscr{D}$ coincides with the injected inventory (§III-A), we re-ran the headline condition (same seeds, CGVD arm) with two *frozen* vocabularies to measure that privilege. With a vocabulary that misses the injected clutter ($\mathscr{D} = \{$ladle, whisk, bowl, cup$\}$), success collapses to 45.5%—statistically indistinguishable from the 43.0% baseline (paired Δ vs. baseline $+2.5$, 95% CI $[-6.6, +11.6]$). With a *generic* 12-category tabletop vocabulary not derived from the injection list (fork, knife, spatula, scissors, ladle, whisk, plate, bowl, cup, mug, pan, pot), success is 76.5%—statistically indistinguishable from the oracle vocabulary (Δ vs. oracle re-execution $+0.5$ $[-4.7, +5.7]$; vs. baseline $+33.5$ $[+22.3, +44.7]$). CGVD's benefit therefore requires vocabulary *coverage* of the clutter, but not knowledge of the exact inventory. Note that the miss condition is not a no-op: the full pipeline still runs—the scene is still edited (mask coverage $\approx$2% vs. $\approx$14% with the oracle vocabulary), the cache still freezes, and the erasure and blending risks remain live—so 45.5% $\approx$ baseline means forfeited benefit, not absent intervention. A near-empty distractor mask is itself a one-line vocabulary-miss detector.

E. Comparison with a Run-Time Intervention Baseline

We compare CGVD against BYOVLA [9], the closest prior run-time observation intervention, at the headline condition (same 10 matched seeds, 200 episodes), reimplementing it from its reference code with three substitutions chosen to remove confounds: GPT-4o replaced by the *same* closed vocabulary $\mathscr{D}$ CGVD receives, GroundingDINO+SAM2 by SAM 3 (shared with CGVD), and Octo's fixed-key stochastic probe by fixed-seed π_0 forward passes. The mechanism is kept faithful,[2] and the probe demonstrably discriminates here (mean 2.0 of 32.1 candidate regions flagged per chunk, range 0–16).

BYOVLA reaches 32.5%—numerically below the 43.0% baseline ($\Delta = -10.5$, 95% CI $[-25.7, +4.7]$, within noise) and far below CGVD's 77.5% (paired $\Delta = -45.0$ $[-56.4, -33.6]$)—while each intervention costs 11.3 ± 2.0 s ($\sim$33 policy forwards): per 4-step action chunk, 11.6 s against CGVD's 0.38 s total (forward plus compositing), a $\approx 30\times$ overhead. We see two causes: per-chunk re-segmentation and re-inpainting makes the background flicker across chunks—Table III's mean-color-fill row shows the policy's sensitivity to fill appearance, the instability CGVD's cached clean plate avoids—and the conservative threshold leaves most of the 18 distractors in view. Two caveats: this is our reimplementation, not the authors' system, and 18 dense distractors is well outside BYOVLA's original regime (a real robot, a handful of distractors); we did not tune its threshold. The fair conclusion is that per-chunk probe-and-inpaint does not scale to dense clutter, not that the method fails in its intended regime.

[2] Per-object Gaussian-blur perturbation (random kernel 15–30); translation-only action deltas with the reference threshold (0.002 m, same physical units); sensitive-region inpainting with the *shared* LaMa inpainter (dilation 10); intervention before every action chunk (4 steps, matching both the π_0 Bridge configuration and the reference implementation's cadence).

F. Latency Analysis

CGVD concentrates its expensive operations (SAM 3, LaMa) on the initialization frame ($t = 0$); for $t > 0$ the system performs image compositing against the cached clean scene. Table IV reports timing measured on an NVIDIA A10G (23 GB) across the evaluation runs; an earlier version of this paper reported a single blended per-step figure, which Table IV replaces with decomposed timings. The policy forward pass is unaffected (318.1 ± 6.6 ms baseline vs. 318.5 ± 5.5 ms under CGVD; one forward per 4-step action chunk), and per-frame compositing adds 15.6 ± 0.4 ms—62 ms per chunk ($\approx$+20% of the forward), i.e. $\approx$+10% episode wall time excluding initialization ($9.11 \rightarrow 10.00$ s in the attribute runs). Initialization is a one-off per-episode cost that grows with the number of segmentation prompts: 0.68 ± 0.29 s in the attribute scenes ($\leq$5 prompts) and $\approx$0.9 s in the 18-distractor scenes (direct measurement over warm episodes; the first episode of a session additionally pays one-time model loading), delaying the first action of each episode. The overhead is modest for the quasi-static tabletop tasks evaluated here; the startup delay remains a real cost for time-critical settings.

TABLE IV

SYSTEM LATENCY, MEASURED ON AN NVIDIA A10G ACROSS THE EVALUATION RUNS (MEAN $\pm$ SD OVER EPISODES; n IN PARENTHESES). CGVD CONCENTRATES SEGMENTATION AND INPAINTING AT t=0; RUNTIME COMPOSITING ADDS $\approx$10% EPISODE WALL TIME.

Quantity	Measured
Policy forward / inference step (one per chunk), baseline	318.1 ± 6.6 ms (500 ep.)
Policy forward / inference step, with CGVD	318.5 ± 5.5 ms (1,100 ep.)
CGVD compositing / frame ($t > 0$)	15.6 ± 0.4 ms (500 ep.)
Initialization ($t = 0$), $\leq$5 prompts	0.68 ± 0.29 s (500 ep.)
Initialization ($t = 0$), 18-distractor scenes	0.91 ± 0.02 s (direct, warm ep.)

V. LIMITATIONS

CGVD's validity envelope is defined by the following assumptions and scope limits.

Static scene. The clean scene is cached at t=0 and α is fixed (§III-F). A moved distractor desynchronizes the cache; worse, an object *entering* the workspace after t=0 can be blended toward the cached background and hidden from the policy while physically present—a hazard, not merely an accuracy loss, outside access-controlled workspaces. Continuous re-segmentation would lift the assumption but is prohibitive at control rate.

Contextually informative clutter. On *put carrot on plate* CGVD never outperforms the baseline and significantly underperforms it at several densities (up to -14.5 pp for GR00T; §IV-B): the intervention is task- and distractor-type-conditional, and deciding *when* to distill is open future work.

Simulation-only evaluation and privileged inputs. No real-robot experiment is reported, and two inputs are simulation-privileged: the renderer robot mask and the per-episode distractor inventory $\mathscr{D}$. SAM 3 recall over $\mathscr{D}$ and LaMa fill fidelity on real textured scenes (the largest single factor in Table III) are unvalidated on hardware; the deployable per-frame SAM 3 robot mask is measured in simulation only (§IV-D), and FK projection on hardware is untested.

Single target instance. Layer 2 retains only the top-scoring component (§III-C); a scene with two genuine target instances would have one inpainted away.

Occlusion fabrication and in-flight blending. Whatever the arm occluded at $t=0$ is synthesized, not observed, in the cached scene (Eq. 6); and since α protects the target only at its $t=0$ location, the *moving* target overlaps previously inpainted regions during execution: in the 200-episode headline re-execution (CGVD arm), the per-frame overlap of the renderer ground-truth target mask with the fixed inpaint mask exceeded 10% of target pixels at some frame in 35% of episodes, exposing the grasped object to partial blending. Finally, initialization delays each episode's first action by 0.7–0.9 s (Table IV).

VI. Conclusion

We introduced CGVD, a training-free inference framework mitigating the Precision-Reasoning Gap in VLA models via language-grounded segmentation and inpainting, without architectural modification. Across more than 20,000 SimplerEnv episodes with π_0 and GR00T, CGVD prevents performance collapse under dense semantic clutter and, in an exploratory compositional-prompt condition, improves attribute adherence, relative to the un-augmented policies; at the headline condition it outperforms a vocabulary-matched BYOVLA reimplementation by 45 pp at $\approx 1/30$th of its per-chunk intervention cost—in a regime outside BYOVLA's design envelope, with its sensitivity threshold untuned. The benefit is conditional: where contextual clutter aids the policy, CGVD never outperforms and often significantly underperforms the baseline. Bounded by static-background and closed-vocabulary assumptions, inference-time visual distillation is thus a practical defense against semantically confusable clutter; future work: real-time mask updating, intrusion detection over compositing regions, open-vocabulary $\mathscr{D}$, and criteria for when to distill.

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